

Rescue Is a Relation, Not a Trait: Construction Dependence, Batch, and the Limits of Certification in a Natural-Variation Screen for Rhizobacterial Drought Rescue in *Arabidopsis*

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September 29, 2026

Abstract

Natural-variation screens for plant responses to beneficial microbes reduce a factorial experiment to a single number per accession and map that number. In a screen of 247 *Arabidopsis thaliana* accessions inoculated with *Pseudomonas simiae* WCS417 under drought, the laboratory derived six families of such numbers (biomass under inoculation, absolute gain, percentage increase, percentage rescue, drought loss, and a binary rescuer label) and mapped each separately. The candidate loci differed by trait, and no accession lost rescue. We reanalyse the screen from its workbook alone. First, every derived trait is an exact function of three cell means (inoculated–drought, mock–drought, mock–non-stress), which we recover to 10^{-10} for all 235 complete accessions; this exposes that one trait family uses no inoculated plants at all and that the published “tolerance” table reports loss. Second, we formalise a *response construction* as a map from cell means to a number that vanishes exactly when inoculation has no effect, and show that the three constructions in use disagree about which accessions rescue best: the top-20% sets overlap by Jaccard 0.32–0.45 in shoot and 0.18–0.27 in root. Under a noise model that includes batch variation, estimated from Col-0 grown in 13 blocks (intraclass correlation 0.28), the shoot disagreement exceeds the resampling ceiling by at most 0.09 while the root disagreement exceeds it by 0.17–0.34, and under plant-level noise alone the constructions certify 63–700 pairs of accessions in *opposite* order. Third, we show that choosing a construction is equivalent to choosing an exponent λ in a scale-invariant family, that λ is estimable from the data, and that the data reject percentage increase ($\lambda = 1$) in both organs ($\hat{\lambda} = -0.26$ [−0.57, −0.02] shoot, −0.04 [−0.26, 0.11] root), after inverting a simulation-calibrated attenuation bias. Within the data-admissible range, construction disagreement falls

below the noise ceiling. Fourth, comparing accessions through four columns (the two mock baselines and the two inoculation responses) with a three-valued verdict (correspond, diverge, decline) whose error rates we calibrate, we find that the screen can certify divergence but essentially never equivalence: the median pair becomes certifiable only at a margin of 1.7–2.8 between-accession standard deviations, and the screen resolves four (shoot) to five (root) tiers of rescue. Of 28 candidate genes, none recurs across trait families, three come from the mock-only trait, and twelve come from a trait whose accession ranking correlates at 0.80 with drought sensitivity. We give a replication design (independent runs dominate plants per cell), a check that separates genuine rescue enhancement from a weaker mock baseline, and a candidate loss-of-rescue accession. The general conclusion is that “rescue” is a relation among cells of a factorial, not a property of an accession, and that the choice among its admissible readings must be made by the data, before mapping, and reported.

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1 Introduction

1.1 The screen

Pseudomonas simiae WCS417 is a model plant-beneficial rhizobacterium that promotes growth, reshapes root architecture and primes systemic resistance in *Arabidopsis* [1; 2; 3]. Part of its effect is linked to the plant’s iron-deficiency machinery [4; 5; 6]. A natural-variation screen under non-stress conditions showed that nearly all accessions respond positively, with large variation in the size of the response [7]. Beneficial microbes are also candidates for improving drought resilience [8], which motivates the screen we reanalyse here: 247 accessions grown with or without WCS417, under non-stress or drought, with shoot and root fresh weight recorded, followed by genome-wide association mapping [9; 10].

The design has eight cells per accession (inoculation × water regime × organ). Association mapping needs one number per accession. The laboratory therefore derived several numbers and mapped each of them:

- biomass under inoculation and drought;
- the absolute gain from inoculation under drought;
- the percentage increase from inoculation under drought;
- a percentage *rescue* of the biomass lost to drought;
- the percentage biomass lost to drought;
- a binary rescuer label.

Different traits returned different candidate loci, and no accession was found to lose rescue altogether. The absence of loss-of-rescue lines made the usual contrast between responders and non-responders unavailable.

Table 1: Contributions, and where each is established.

Contribution	Where
Exact recovery of the unobserved cell means from derived traits; audit of labels	§4.1
Admissible response constructions; scale invariance; the λ family	§3.1
Heterogeneity and detectability of rescue	§4.2
Construction dependence, measured against three noise models	§4.3
A data-driven choice of construction with bias correction	§4.4
Four-column comparison with calibrated three-valued verdicts	§4.5
Batch estimated from an internal control	§4.6
Consequences: candidate triage, mutant checks, design	§5

1.2 The question

Two questions follow and they are usually conflated. The first is biological: which host loci modulate WCS417-mediated drought rescue? The second is prior to it and methodological: *which number is the rescue of an accession?* A GWAS answers the first only after the second has been answered, and the second is not answered by the data unless one asks it explicitly. The candidate lists differ between traits. That difference is either noise, which would argue for pooling, or it is systematic, which would mean the traits measure different things and at most one of them measures rescue.

Reducing a factorial to a ratio also has a long history of producing artefacts. Ratios of noisy quantities correlate spuriously with their own denominators [11; 12]. Drought indices built from stressed and unstressed yield are well known to conflate potential with tolerance [13; 14]. Gene-by-environment interaction is the rule in *Arabidopsis*, not the exception [15; 16]. None of this is new. What is missing is a procedure that, given a screen like this one, says which readings of the phenotype are admissible, lets the data choose among them, measures how much the choice matters, and says how much any comparison between accessions can be trusted.

1.3 What this paper does

We build that procedure from first principles and apply it to the screen. Table 1 lists the contributions.

1.4 What this paper does not do

- (1) We do not rerun the association scan. The workbook contains no genotypes, and we did not retrieve them. Every statement about candidate loci concerns *which trait* a candidate came from, not whether it is real.

- (2) Per-plant values exist for one cell of the design only. For the other three cells we recover exact means but not their spread, so the noise in those cells is modelled, not measured (Assumption 3.1). Every result that depends on this is reported under several noise models.
- (3) We do not claim that any accession fails to be rescued. We identify accessions whose rescue is statistically compatible with zero; that is a reason to retest them, not a finding.
- (4) We use no data beyond the workbook, the laboratory’s slide deck, and the voice-recorded description of the design by the experimenter.

2 The screen as received

Materials. The workbook contains six sheets:

- *Biomass data*: per-plant rows;
- *easyGWAS input*: one row per accession with 38 derived traits and two binary labels;
- *easyGWAS output*: 40 candidate rows covering 28 genes;
- *Accessions*: identifier, name, country and coordinates;
- *Table 1* and *Table 2*: summaries prepared for a manuscript.

Accessions are keyed by their 1001 Genomes identifiers [17].

What was measured and what was kept. The per-plant sheet holds 1,841 plants. *All* of them belong to one cell: WCS417-inoculated, drought. There are 243 accessions with 7 plants each, three accessions (7394, 9807, 9874) with 14 plants in two blocks, and Col-0 (6909) with 98 plants in 14 blocks, one of which has no recorded weights. The mock plants and the non-stress plants appear only through per-accession averages and medians embedded in the derived traits. Primary root length is recorded for 1,042 plants.

Analysis set. We exclude the four repeatedly grown lines, because their derived traits were computed from inconsistent plant subsets (§4.1). We also exclude eight accessions with missing mock values. The analysis set is 235 accessions. Throughout, “shoot” and “root” refer to fresh weight in mg.

Collection sites. The 235 accessions come from 30 countries, led by Spain (66), Germany (44), Russia (18) and France (17). They are not a random draw from the species. They are the collection that was available in the laboratory, which had originally been assembled for a study of annual precipitation. We return to this in §4.7.

3 Theory

3.1 Cell means and response constructions

For an accession a and an organ, write

$$W_a, \quad M_{D,a}, \quad M_{N,a}, \quad W_{N,a}$$

for the mean fresh weights under inoculation–drought, mock–drought, mock–non-stress and inoculation–non-stress respectively. The question “how much does WCS417 rescue a under drought?” is a question about a *relation* among these cells, and there is more than one way to turn it into a number.

Definition 3.1 (Response construction). A *response construction* is a function $f : (W, M_D, M_N) \mapsto \mathbb{R}$, defined for $M_N > M_D > 0$ and $W > 0$, with two properties:

- (i) *null calibration*: $f = 0$ if and only if $W = M_D$;
- (ii) *monotonicity*: f is strictly increasing in W .

It is *scale invariant* if, in addition, $f(sW, sM_D, sM_N) = f(W, M_D, M_N)$ for every $s > 0$.

Scale invariance encodes a specific biological requirement. Accessions differ in overall vigour, and a vigorous accession has larger fresh weights in every cell. If rescue is a property of the interaction and not of vigour, multiplying all four cells of an accession by the same factor should leave its rescue unchanged. The three constructions used in the screen are:

$$g = W - M_D, \tag{1}$$

$$\iota = W/M_D - 1, \tag{2}$$

$$\varrho = (W - M_D)/(M_N - M_D). \tag{3}$$

Here g is the absolute gain, ι the proportional increase over the mock plant under the same drought, and ϱ the fraction of the drought loss that inoculation restores.

Proposition 3.1 (Which constructions are scale invariant). *All three constructions are admissible. Of them, ι and ϱ are scale invariant and g is not.*

Proof. Null calibration and monotonicity are immediate in each case, because $M_D > 0$ and $M_N - M_D > 0$. Under $(W, M_D, M_N) \mapsto s(W, M_D, M_N)$, $g \mapsto sg$, whereas the numerator and the denominator of $\iota + 1$ and of ϱ both scale by s . $\square$

The two scale-invariant constructions are members of one family.

Definition 3.2 (The λ family). For $\lambda \in \mathbb{R}$ let

$$f_\lambda(W, M_D, M_N) = \frac{W - M_D}{M_D^\lambda (M_N - M_D)^{1-\lambda}}. \tag{4}$$

Then $f_1 = \iota$ and $f_0 = \varrho$.

The denominator is homogeneous of degree one, so every f_λ is scale invariant. The exponent λ says which baseline the gain is measured against: the size of the stressed mock plant ($\lambda = 1$), the size of the drought loss ($\lambda = 0$), or a geometric compromise. The following proposition makes precise the sense in which choosing λ is choosing a mechanism.

Proposition 3.2 (The construction encodes a null mechanism). *Suppose the gains are generated as*

$$W_a - M_{D,a} = \theta_a M_{D,a}^{\lambda^*} (M_{N,a} - M_{D,a})^{1-\lambda^*}, \quad (5)$$

with a rescue capacity $\theta_a > 0$ that is statistically independent of the baselines $(M_{D,a}, M_{N,a})$. Then:

- (i) $f_{\lambda^*}(a) = \theta_a$ is independent of both baselines;
- (ii) for any $\lambda \neq \lambda^*$,

$$f_\lambda(a) = \theta_a (M_{D,a}/(M_{N,a} - M_{D,a}))^{\lambda^* - \lambda},$$

which depends on the baselines unless $M_D/(M_N - M_D)$ is constant across accessions.

Proof. Substitute (5) into (4). $\square$

The ratio $M_D/(M_N - M_D)$ is a monotone transform of drought sensitivity: it is small when drought removes most of the biomass. So the wrong λ loads the phenotype onto drought sensitivity, and the sign of $\lambda^* - \lambda$ sets the direction of the loading. Percentage increase is the right construction only if WCS417 multiplies the stressed biomass by an accession-specific factor. Rescue is the right one only if WCS417 restores an accession-specific fraction of the biomass lost. Nothing in the definition of “rescue” decides between these; the data can.

Estimating λ . Taking logarithms of (5) gives a linear model

$$\log(W - M_D) = b_0 + b_1 \log M_D + b_2 \log(M_N - M_D) + \varepsilon, \quad (6)$$

in which the mechanism predicts $b_1 = \lambda^*$ and $b_1 + b_2 = 1$. The second of these is scale invariance of the mechanism itself, and is testable. Because M_D enters both sides with measurement error, the ordinary least-squares b_1 is biased [18]. We estimate the bias by simulation at the observed noise level and invert it (§3.6).

3.2 Receiver relativity

Let $y_a = f(a)$ be the response of accession a under a fixed construction, and se_a its standard error. If WCS417 had one effect on all accessions, the y_a would differ only by noise.

Definition 3.3 (Heterogeneity). With inverse-variance weights $w_a = se_a^{-2}$ and weighted mean $\bar{y}$, Cochran’s statistic [19]

$$Q = \sum_a w_a (y_a - \bar{y})^2$$

is χ_{n-1}^2 under a common effect. The excess $I^2 = (Q - (n-1))/Q$ [20] and the reliability

$$\mathcal{R} = \frac{\tau^2}{\tau^2 + se^2}, \quad \tau^2 = \text{Var}(y) - se^2,$$

measure how much of the observed spread is between accessions.

$\mathcal{R}$ plays the role of an entry-mean repeatability [21; 22]: it bounds the variance that any locus could explain. A large Q means the effect of the strain has no single value. It has one value per accession, each a property of the pair (strain, accession), and the object of a mapping study is that indexed set, not a number attached to the strain.

3.3 Construction independence

Let $\mathcal{F}$ be a set of admissible constructions and let V_f be a verdict computed from construction f . Examples of verdicts are the set of the top k accessions, or the certified order of a pair.

Definition 3.4 (Construction independence). A verdict is *construction independent over $\mathcal{F}$ at noise level $\mathcal{M}$* if, for all $f, g \in \mathcal{F}$, the disagreement between V_f and V_g on the observed data does not exceed the disagreement between V_f on the observed data and V_f on data resampled under the noise model $\mathcal{M}$ (95th percentile).

The comparison is the substance of the definition. Two constructions can disagree because they read the same truth differently, or because each is noisy. Only the first is a property of the constructions, and it is detectable only as disagreement beyond what resampling produces.

A stronger failure than disagreement is contradiction.

Definition 3.5 (Certified order; decisive contradiction). Construction f *certifies* $a \succ b$ if

$$f(a) - f(b) - z_\alpha \sqrt{se_a^2 + se_b^2} > \delta.$$

Two constructions are in *decisive contradiction* on a pair if one certifies $a \succ b$ and the other certifies $b \succ a$.

3.4 Four columns and three-valued verdicts

Comparing two accessions by their response alone discards what they share. Two accessions can have the same response for different reasons, or different responses from indistinguishable starting points. We therefore compare a pair (a, b) through four columns:

- two *baseline* columns, the mock phenotype vector $(\log M_D, \log M_N)$ of a and of b ;
- two *response* columns, $f(a)$ and $f(b)$.

Each pair of columns is compared by a verdict with three values.

Definition 3.6 (Three-valued verdict). For an observed difference d with standard error s and a margin $\delta > 0$, set

$$V(d, s; \delta) = \begin{cases} \text{C (correspond)} & |d| + z_\alpha s < \delta, \\ \text{D (diverge)} & |d| - z_\alpha s > \delta, \\ \text{U (decline)} & \text{otherwise,} \end{cases}$$

with z_α the one-sided $1 - \alpha$ normal quantile.

For a two-component baseline, the verdict is C if both components are C, D if either is D, and U otherwise (intersection-union).

C is the two one-sided tests rule for equivalence [23; 24]. D is its mirror image: a certification that the difference exceeds the margin. U is not a failure of the procedure. It is the correct report when the data do not settle the question, and a procedure without it must return a confident answer for every pair.

Proposition 3.3 (Error control). *Let $d \sim \mathcal{N}(\Delta, s^2)$ with s known. Then*

- (i) $\Pr[V = C] \leq \alpha$ whenever $|\Delta| \geq \delta$;
- (ii) $\Pr[V = D] \leq \alpha + \Phi(-2\delta/s - z_\alpha)$ whenever $|\Delta| \leq \delta$.

Proof. (i) Take $\Delta \geq \delta$ without loss of generality. C requires $d < \delta - z_\alpha s$, which has probability $\Phi((\delta - \Delta)/s - z_\alpha) \leq \Phi(-z_\alpha) = \alpha$.

(ii) D requires $d > \delta + z_\alpha s$ or $d < -\delta - z_\alpha s$. For $|\Delta| \leq \delta$, the first has probability at most α , attained at $\Delta = \delta$. The second is then $\Phi(-2\delta/s - z_\alpha)$ at the same Δ , and smaller elsewhere. $\square$

The four columns define a 3×3 table of (baseline, response) verdicts. Two cells of it carry the diagnostic content:

- (C, D): *false friends*, indistinguishable in mock but different in response;
- (D, C): *convergent* pairs, different in mock but equivalent in response.

A comparison that reads only the response columns cannot tell a false friend from an ordinary divergent pair. A comparison that reads only the biomass under inoculation mixes the columns altogether.

Definition 3.7 (Minimum certifiable margin). For a pair, $\delta_{\min} = |d| + z_\alpha s$. This is the smallest margin at which the pair would be certified C.

The distribution of $\delta_{\min}$ over pairs, expressed in units of the between-accession standard deviation, is a design-level quantity. It says how coarse a notion of “the same rescue” the screen can support at all. If the median pair needs a margin larger than the population spread, the screen cannot certify any meaningful equivalence.

Definition 3.8 (Resolution depth). Let $a \succ b$ mean certified order at margin 0. The *resolution depth* is the length of the longest chain $a_1 \succ a_2 \succ \dots \succ a_L$. The *tier* of an accession is the length of the longest chain ending at it from below.

Resolution depth counts how many levels of rescue the screen can tell apart, which matters more to a follow-up experiment than how many accessions it ranks.

3.5 Noise models

Assumption 3.1 (Noise). Cell means are perturbed multiplicatively and log-normally. The plant coefficient of variation c_p is the median within-accession coefficient of variation of the observed inoculated-drought plants, and it is assumed to hold in the three unobserved cells. The batch coefficient of variation c_b is the between-block excess variance of Col-0 block means (§4.6). Batch effects are independent across cells.

The last clause is conservative. If an accession’s mock and inoculated plants shared a tray, part of the batch effect would cancel in $W - M_D$. We therefore bracket the truth with four models:

M1 resample the observed inoculated plants; mock means fixed.

M2 *M1*, plus mock means perturbed with coefficient of variation $c_p/\sqrt{7}$.

M3 *M2*, plus a batch factor on every cell. This is the reference model.

M4 *M3* with the mock noise multiplied by 1.5.

We read *M2* and *M3* as lower and upper bounds on realistic noise.

3.6 Estimating λ without bias

Uncertainty. We fit (6) by ordinary least squares and bootstrap [25] jointly over accessions (case resampling) and noise (a fresh *M3* draw per replicate), with 1,000 replicates.

Bias. To calibrate, we simulate synthetic screens for $\lambda^* \in \{-0.75, -0.5, \dots, 1\}$:

- the observed (M_D, M_N) are taken as the true baselines;
- $\log \theta_a$ is normal with the residual standard deviation of the observed fit;
- gains are scaled to the observed median, and *M3* noise is added.

We run 200 replicates per value of λ^* . The mean estimate is monotone in λ^* . We invert that curve to obtain a corrected $\hat{\lambda}$, and apply the same inversion to the bootstrap interval.

4 Results

4.1 Audit of the workbook

Recovering the cell means (Figure 1). The absolute gain under drought and the percentage loss under

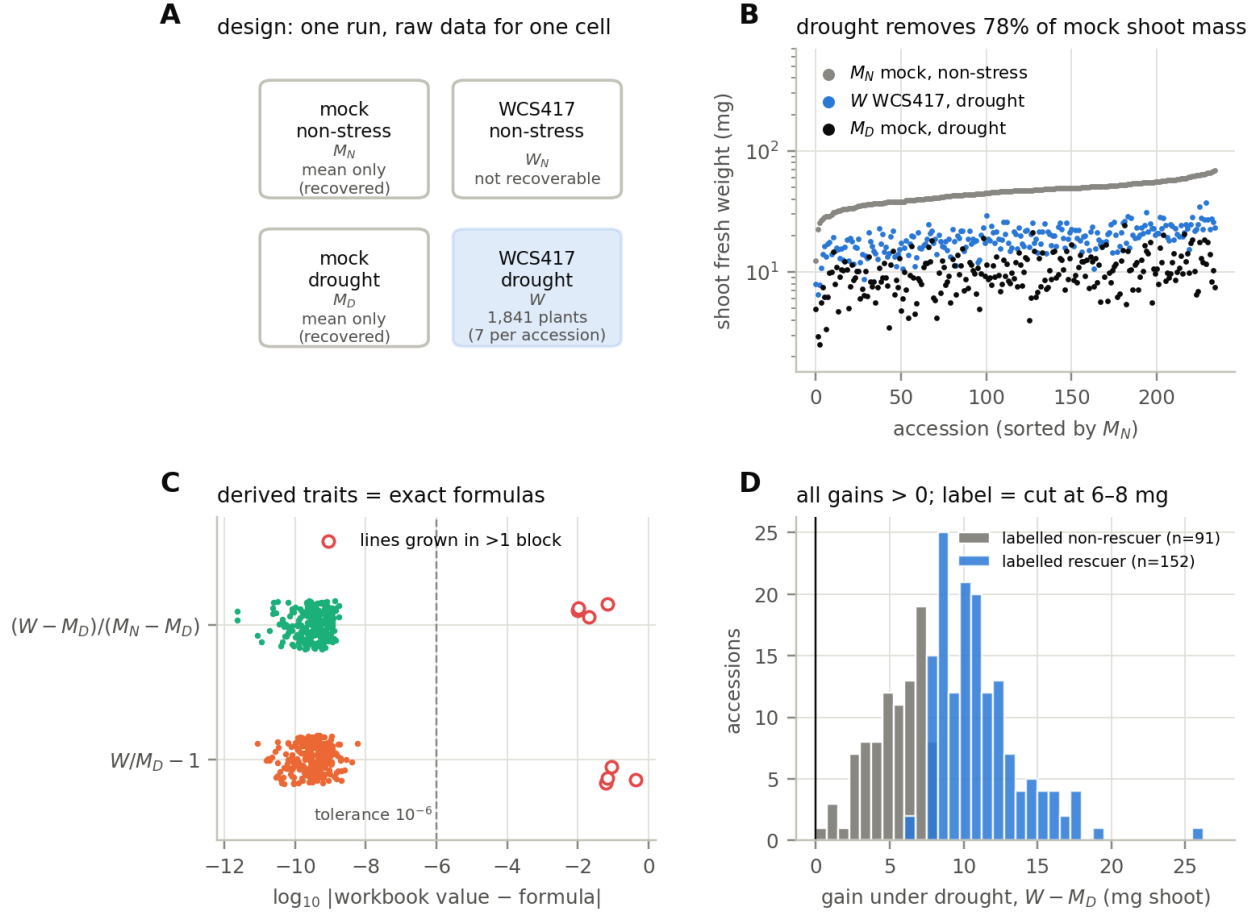


Figure 1: The workbook determines three of four cell means exactly, and one trait family uses no inoculated plants. (A) Design and data availability. Per-plant weights exist only for WCS417–drought (1,841 plants). Mock means are recovered from the derived traits (§4.1). WCS417–non-stress cannot be recovered, because the one trait that involves it admits two readings. (B) Recovered shoot cell means for 235 accessions, sorted by M_N , on a log scale. Drought removes a median 78% of mock shoot biomass (84% of root). Inoculation recovers a varying part of it. (C) Absolute residual between each workbook trait and its closed form, on a log scale. For all 235 accessions in the analysis set, the rescue and increase traits equal (3) and (2) to rounding error (median 4×10^{-10}). The four exceptions (red) are the lines grown in more than one block, whose traits mix plant subsets. (D) Distribution of the shoot gain $W - M_D$ by the workbook’s binary label. No accession has a negative gain: the smallest is 0.73 mg. The label is a cut between 6.1 and 7.8 mg, not a loss of rescue.

drought determine the mock means exactly:

$$M_D = W - \text{gain}_D, \quad M_N = M_D / (1 - \text{loss}).$$

Because W is the mean of the recorded plants (maximum discrepancy 4×10^{-9}), both mock cells follow for every accession.

Checking the recovery. Two traits not used in the recovery serve as independent checks:

- the workbook’s percentage rescue equals (3) for all 235 accessions (median absolute residual 3.8×10^{-10} shoot, 5.9×10^{-10} root);
- the percentage increase equals (2) for 239 of 247.

The only failures of either identity are the four repeatedly grown lines, with residuals 0.007–0.13. Their traits were computed from differing plant subsets, and we exclude them.

The unrecoverable cell. The “gain under non-stress” trait is consistent to rounding error with two different definitions, $W_N - M_N$ and $W - W_N$. W_N is therefore not identifiable from the workbook, and no analysis below uses it.

Three corrections to the laboratory’s reading.

- The “loss under drought” trait, $1 - M_D/M_N$, uses mock plants only. It measures drought sensitivity, not rescue.
- The “drought tolerance” columns of the laboratory’s Table 2 are identical to that loss for 239 of 240 accessions. A larger value therefore means *less* tolerance, the opposite of what the label says.
- The binary rescuer label is a threshold on a positive gain (Figure 1D). Labelled non-rescuers gain 0.73–7.75 mg shoot; labelled rescuers gain 6.09–25.6 mg. Every accession gains.

Table 1 of the laboratory reports the drought condition: its mock and WCS417 columns match M_D and W with $r = 0.96$ and 0.97 ($n = 243$).

4.2 Heterogeneity and detectability

Noise parameters. The plant coefficient of variation is $c_p = 0.121$ (shoot) and 0.156 (root). Col-0 gives $c_b = 0.081$ and 0.093 .

Heterogeneity. Inoculation effects differ between accessions far beyond noise, for every construction, organ and noise model (Table 2). Under M3 the shoot Q ranges from 396 to 877 on 234 degrees of freedom. So the effect of WCS417 under drought is receiver-relative in the sense of §3.2: it has no single value.

Table 2: Heterogeneity of the inoculation effect across 235 accessions ($df = 234$). $\mathcal{R}$ is the reliability. All $p < 10^{-9}$.

Organ	Construction	Q (M2)	Q (M3)	$\mathcal{R}$ (M3)
Shoot	gain	3,422	757	0.66
	increase	3,692	877	0.78
	rescue	1,710	396	0.35
Root	gain	3,869	931	0.68
	increase	3,879	1,048	0.83
	rescue	1,899	516	0.48

Reliability. Reliability differs sharply between constructions. Rescue, a ratio of two differences, is the noisiest. Under M3 its between-accession standard deviation is smaller than its per-accession standard error (ratio 0.77 in shoot). A GWAS on shoot rescue from this screen maps a phenotype of which about a third of the variance is between accessions.

Could a loss of rescue have been detected? Under M3 the median shoot gain is 4.5 standard errors from zero, and 92% of accessions have intervals that exclude zero (96% in root). The minimum gain detectable with 80% power is 5.8 mg shoot, against an observed median of about 9 mg. A complete loss of rescue in most genetic backgrounds would therefore have been visible.

Retest candidates. Eighteen shoot intervals and nine root intervals include zero under M3. Under M2 the numbers are six (shoot) and one (root). Per-1 has the smallest gain in both organs (0.73 mg shoot, 0.27 mg root) and is compatible with zero under every model. It is the natural first candidate for retesting as a loss-of-rescue line.

4.3 Construction dependence

Correlations (Figure 3A). The three constructions are correlated but far from interchangeable. The pairwise Spearman correlations [26] are:

- gain with increase 0.79 and gain with rescue 0.78, but increase with rescue only 0.66 (shoot);
- 0.59, 0.61 and 0.41 respectively in root.

Why they disagree (Figure 3A, B). Each construction carries a different nuisance, as Proposition 3.2 predicts.

- Percentage increase ranks accessions almost as mock-only drought loss does ($\rho = 0.80$ in both organs). A drought-sensitive accession has a small M_D and hence a large ratio. This is the spurious ratio correlation of Pearson [11], reappearing as a phenotype.
- Gain in milligrams tracks size ($\rho = 0.60$ with W).
- Rescue is loaded on neither beyond $|\rho| = 0.41$, but it is the least reliable (Table 2).

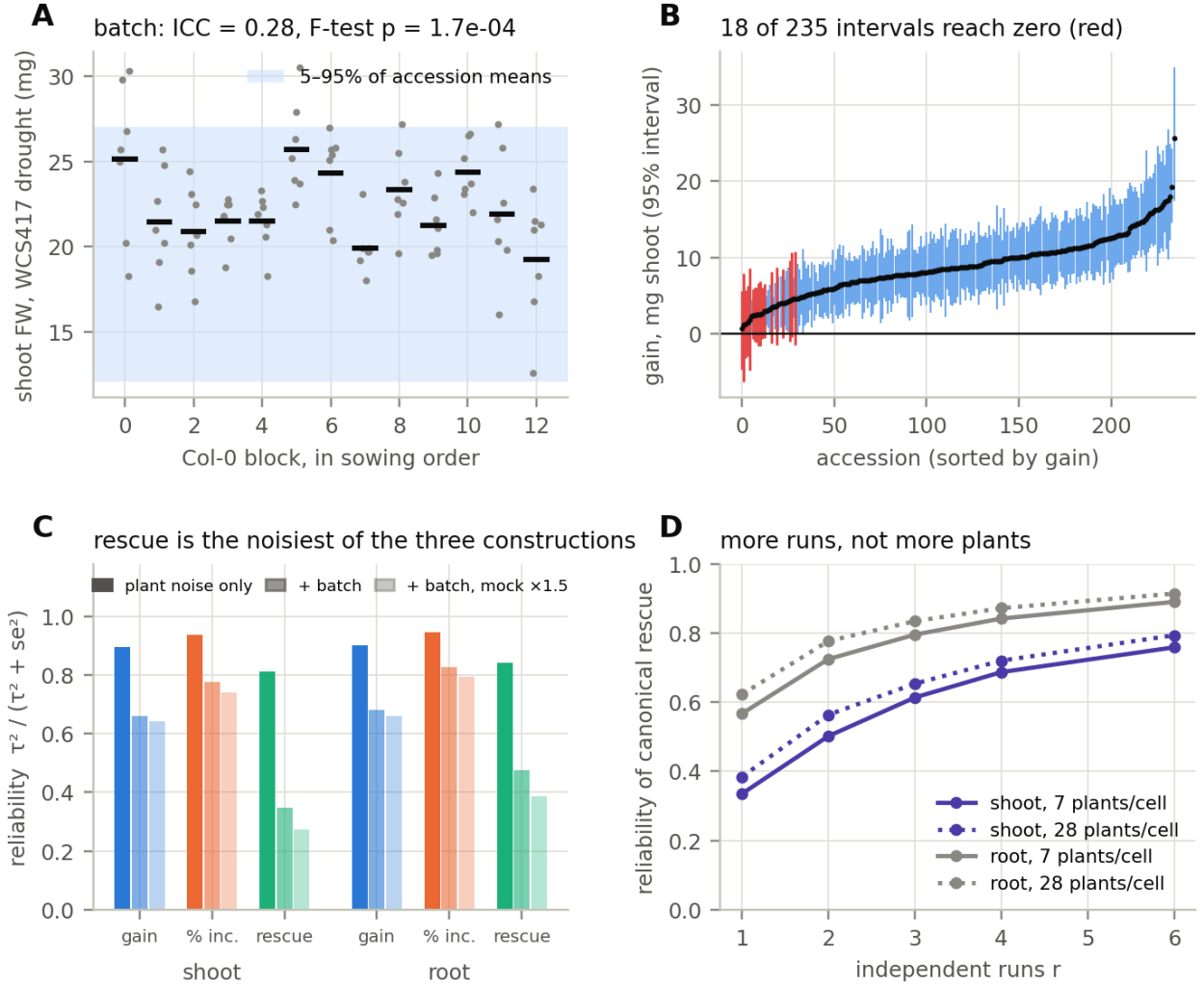


Figure 2: Batch is a first-order component of noise, rescue is the least reliable construction, and replication should add runs. (A) Col-0 was grown in 13 usable blocks of 7 plants spread through the screen. Block means (bars) differ beyond plant noise ($F = 3.76$, $p = 1.7 \times 10^{-4}$, ICC = 0.28). The spread of the block means is 0.43 of the between-accession spread, and Col-0 alone moves from the 54th to the 92nd percentile of accessions. (B) Shoot gain with 95% intervals under the reference noise model (M3), sorted. Eighteen of 235 intervals include zero (red). Under plant noise alone (M2) six do. Per-1 is compatible with zero in shoot and root under every model. (C) Reliability $\mathcal{R}$ (Definition 3.3) by construction and noise model. Adding batch noise lowers the reliability of rescue to 0.35 (shoot) and 0.48 (root); the other two constructions remain between 0.66 and 0.83. Heterogeneity is significant for every construction, organ and model (Table 2). (D) Reliability of the canonical construction as a function of independent runs r , for 7 and 28 plants per cell. Quadrupling the plants moves reliability by at most 0.05; a second run moves it by 0.15–0.17.

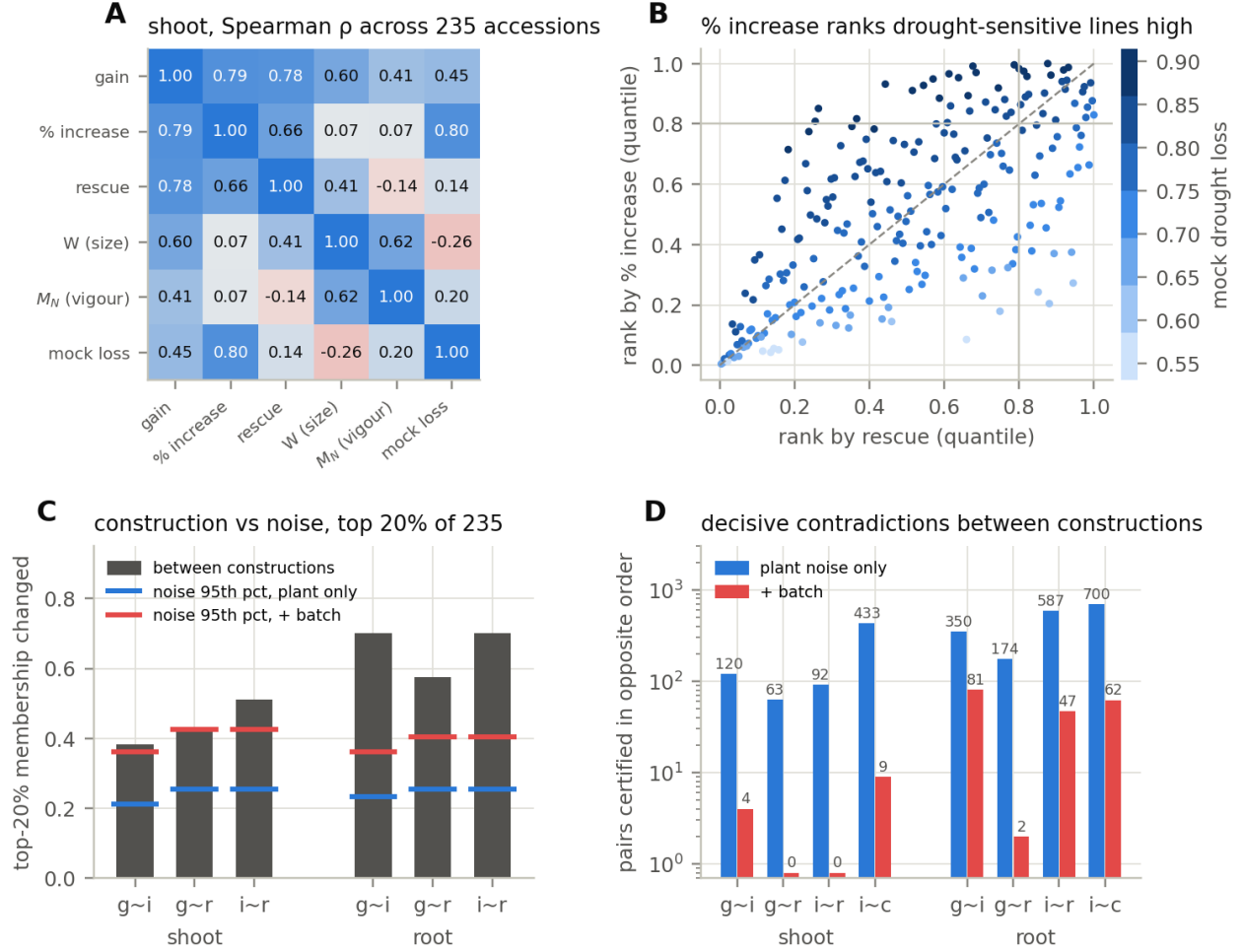


Figure 3: The three constructions in use disagree about which accessions rescue best, and sometimes certify opposite orders. (A) Spearman correlations across 235 accessions (shoot) between the constructions and three nuisance quantities: size under inoculation W , vigour M_N , and mock-only drought loss. Percentage increase is nearly uncorrelated with size but correlates at 0.80 with drought loss. Gain correlates at 0.60 with size. Rescue is the least loaded on either. (B) Rank by percentage increase against rank by rescue, coloured by drought loss. Accessions far above the diagonal are drought-sensitive: their small M_D inflates the ratio. (C) Fraction of the top 20% (47 accessions) that changes between two constructions (bars), against the 95th percentile of the same fraction under resampling of one construction: plant noise (blue) and plant + batch noise (red). In root every pair exceeds even the batch-inclusive ceiling (0.57–0.70 against 0.36–0.40). (D) Pairs of accessions certified in *opposite* order by two constructions (Definition 3.5, $\delta = 0.5$ SD). Under plant noise these number 63–433 in shoot and 174–700 in root. With batch noise they fall to 0–81.

Top sets (Figure 3C). The top-20% sets overlap by Jaccard index [27] 0.32–0.45 in shoot and 0.18–0.27 in root. Only 15 of 47 shoot accessions, and 7 of 47 root accessions, are in the top fifth under all three constructions. At the top 10% no root accession is shared by all three.

Against the noise ceilings of Definition 3.4:

- *Shoot.* Construction disagreement exceeds the plant-noise ceiling for every pair (e.g. 0.51 against 0.26 for increase–rescue) but exceeds the batch-inclusive ceiling only marginally (0.51 against 0.43; excess 0.00–0.09 across the three pairs).
- *Root.* It exceeds both ceilings for every pair, by 0.17–0.34.

So construction independence fails in root under every noise model. In shoot it fails clearly without batch noise. With batch noise the construction question is nearly swamped, but only because the screen itself becomes too noisy to rank.

Contradictions (Figure 3D). Disagreement is not only a matter of which verdicts are reached; some verdicts are reversed. Under plant noise, percentage increase and the canonical construction (§4.4) certify 433 shoot pairs and 700 root pairs in opposite order. Under the batch-inclusive model 9 and 62 such contradictions survive. A contradiction is not a disagreement about resolution. It is two admissible readings of the same data asserting, each at the 5% level, incompatible facts.

4.4 Letting the data choose

The estimate (Figure 4). The uncorrected fit of (6) gives $b_1 = -0.34 [-0.63, -0.12]$ in shoot and $-0.09 [-0.30, 0.06]$ in root. Simulation shows the estimator is attenuated by 0.10 (shoot) and 0.04 (root) near $\lambda^* = 0$. Correcting gives

$$\hat{\lambda}_{\text{shoot}} = -0.26 [-0.57, -0.02], \quad \hat{\lambda}_{\text{root}} = -0.04 [-0.26, 0.11].$$

Percentage increase ($\lambda = 1$) is excluded with a wide margin in both organs. Rescue ($\lambda = 0$) lies inside the root interval and just beyond the upper end of the shoot interval (-0.02). We adopt $f_{\hat{\lambda}}$, per organ, as the *canonical* construction. For practical purposes it is rescue: the two correlate at ≥ 0.88 in tier assignment and never certify opposite orders.

Scale invariance of the mechanism. The mechanism (5) predicts $b_1 + b_2 = 1$. The estimate is attenuated too: under exact invariance, simulation gives 0.81 ± 0.17 (shoot) and 0.86 ± 0.12 (root).

- *Shoot.* The observed 0.61 is within 1.2 simulation standard deviations, so it is compatible with scale invariance.
- *Root.* The observed 0.48 is 3.2 standard deviations below, so it is not.

Larger root systems gain proportionally less. Correspondingly, no member of the λ family removes the root loading on vigour: at $\lambda \approx 0$, $\rho(f_{\lambda}, M_N) = -0.34$. In root, the vigour dependence should be handled in the association model, by entering $\log M_N$ as a covariate, rather than by the phenotype.

Does canonicalisation restore construction independence? Within the data-admissible interval of λ , the top-20% change between the two interval ends is 0.32 (shoot) and 0.30 (root). Both are below the batch-inclusive noise ceilings of 0.43 and 0.40. The disagreement between the constructions in use (0.51 and 0.70) exceeds them. *Once the data are allowed to choose, the residual choice no longer changes the verdict more than noise does.*

4.5 Four columns and the limits of certification

Calibration (Figure 5A). Across five noise levels, false correspondence is at most 0.051 and false divergence at most 0.054, as Proposition 3.3 requires. The verdict does what it claims.

Equivalence is out of reach (Figure 5B). This screen’s ratio of pair standard error to a margin of 0.5 between-accession SD is 2.0. At that ratio Definition 3.6 can almost never return C. The minimum-certifiable-margin distribution makes the limitation exact:

- under M3 the median pair certifies as equivalent only at 2.75 SD (shoot) or 2.41 SD (root), a margin wider than the spread of the population;
- even under plant noise alone the medians are 1.80 and 1.69 SD, and only 14–16% of pairs certify below 1 SD.

The screen can say that two accessions differ. It cannot say that two accessions rescue equally.

The four-column table (Figure 5C). Under plant noise, the shoot baseline columns certify 2,892 pairs as equivalent at a 30% margin. Of these:

- 818 are false friends: certified *different* in response despite indistinguishable mock phenotypes. The most extreme include IP-Mur-0 with Lu3-30 (canonical rescue 0.05 against 0.31) and Neo-6 with Uk-1.
- 20 pairs are convergent (root: 43).

False friends are the informative contrasts for mapping. Their difference in response cannot be attributed to a difference in baseline growth or drought sensitivity. Under M3 only 18 shoot false friends survive, and no convergent pair does, because C is unreachable in the response columns.

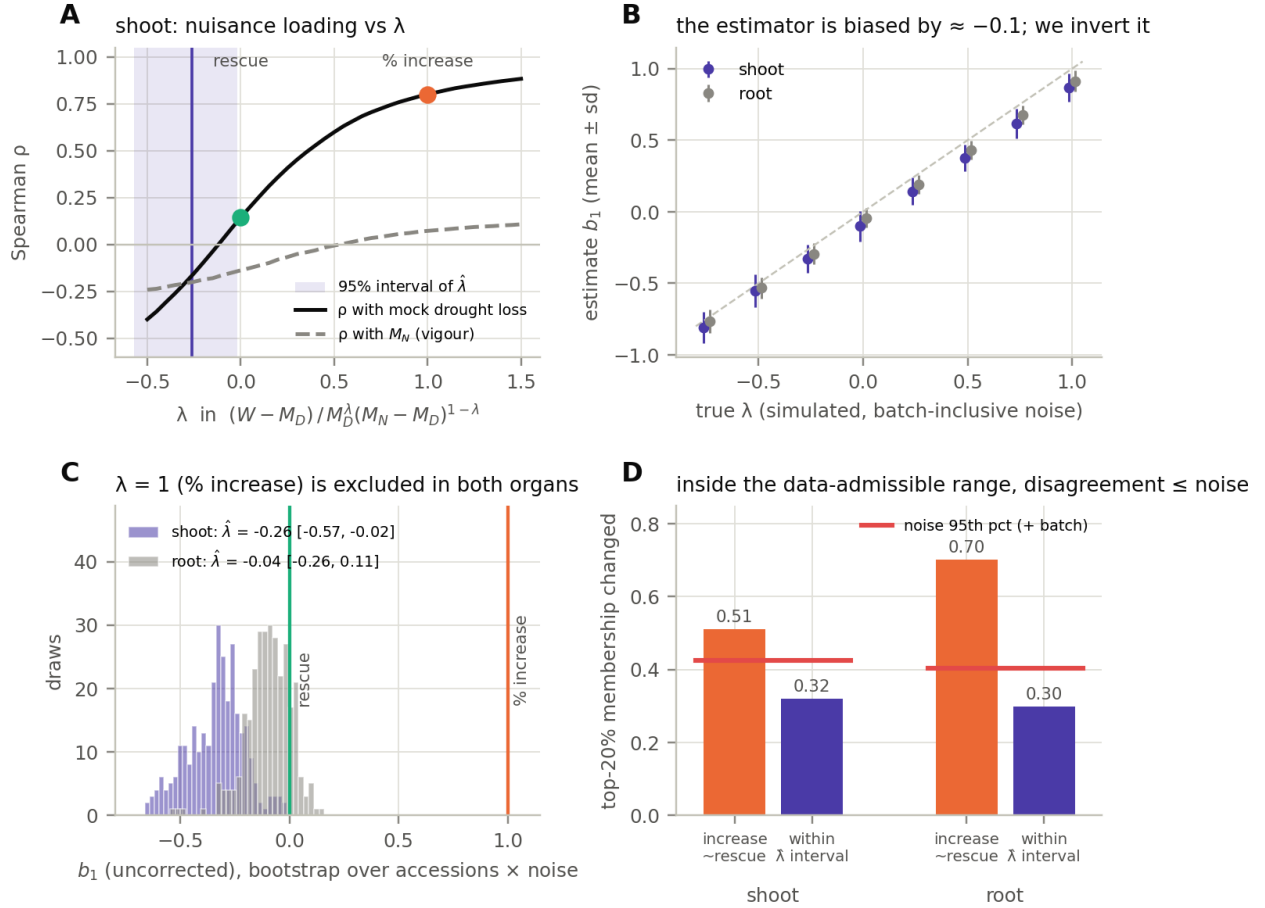


Figure 4: The data choose the construction: percentage increase is excluded, and within the admissible range the disagreement falls below noise. (A) Spearman loading of f_λ on mock drought loss (solid) and on vigour M_N (dashed) as λ varies, shoot. At $\lambda = 1$ (percentage increase) the loading on drought loss is 0.80. Near $\lambda = 0$ (rescue) both loadings are small. The shaded band is the 95% interval of the corrected $\hat{\lambda}$. (B) Recovery of a known λ^* from simulated screens with batch-inclusive noise (200 replicates per point). The least-squares slope is biased downward by 0.02–0.13, and more so for larger λ^* . We invert the curve. (C) Bootstrap distribution of the uncorrected slope b_1 over accessions and noise. After correction, $\hat{\lambda} = -0.26$ $[-0.57, -0.02]$ (shoot) and -0.04 $[-0.26, 0.11]$ (root). $\lambda = 1$ lies far outside both. (D) Top-20% change between increase and rescue (orange) compared to change between the two ends of the $\hat{\lambda}$ interval (violet), against the batch-inclusive noise ceiling (red). Restricting to data-admissible constructions brings disagreement to 0.32 (shoot) and 0.30 (root), below the ceilings of 0.43 and 0.40.

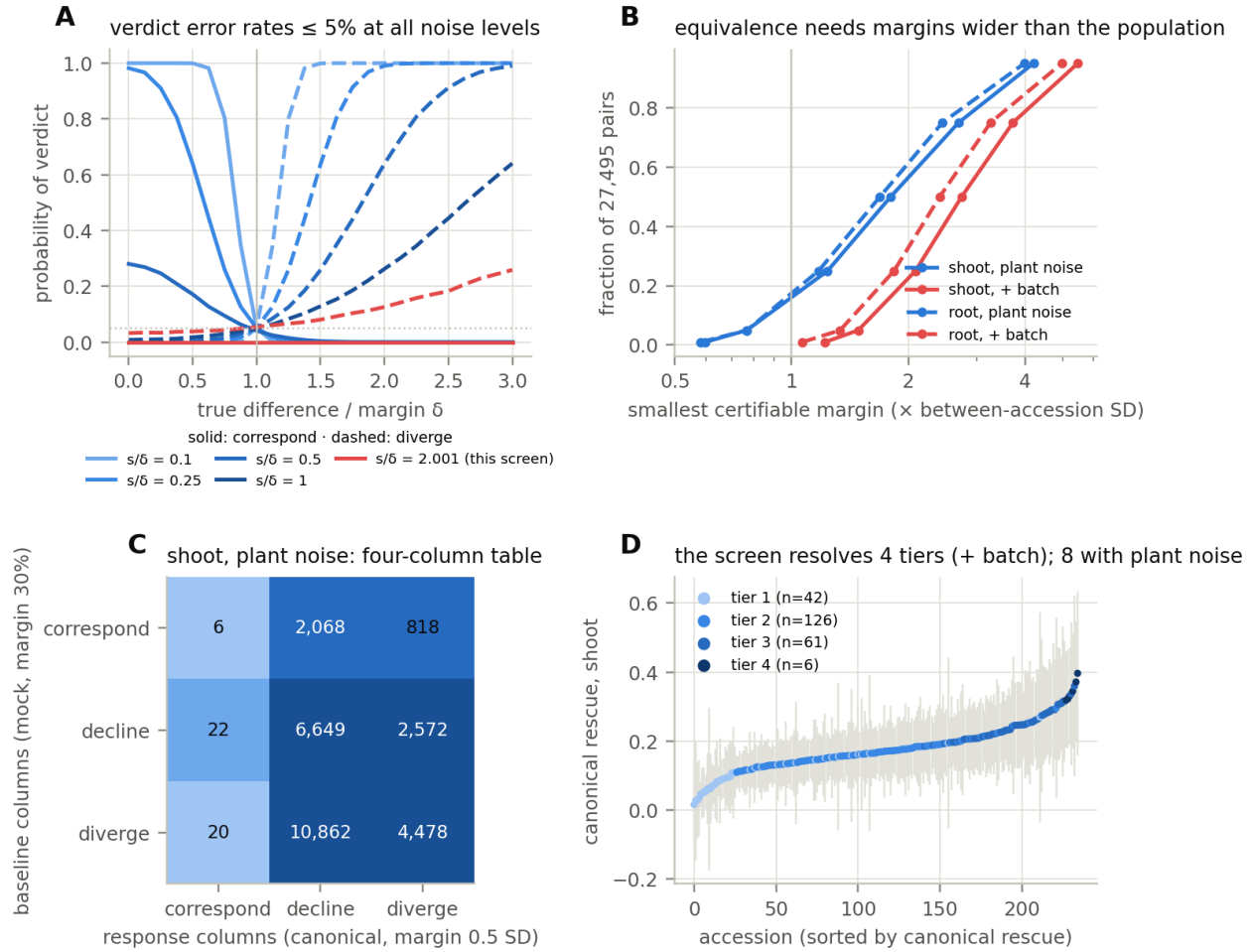


Figure 5: The verdict is calibrated, and the screen can certify divergence but almost never equivalence. (A) Operating characteristics of Definition 3.6 by simulation, at five ratios of pair standard error to margin. Solid lines: probability of C; dashed lines: probability of D. False correspondence beyond the margin never exceeds 0.051 (Monte Carlo error 0.001). False divergence within the margin never exceeds 0.054, which is the bound of Proposition 3.3(ii). At this screen's ratio (red, $s/\delta = 2$), C is unreachable. (B) Cumulative distribution over all 27,495 pairs of the minimum certifiable margin (Definition 3.7), in units of the between-accession standard deviation. With plant noise the median pair needs 1.7–1.8 SD; with batch 2.4–2.8 SD. Only 0.07% (shoot) and 0.55% (root) of pairs certify below 1 SD under M3. (C) The four-column table for shoot under plant noise (baseline margin 30%, response margin 0.5 SD). There are 818 false friends (C, D) and 20 convergent pairs (D, C). Most pairs decline in at least one column. (D) Canonical rescue with 95% intervals, coloured by resolution tier (Definition 3.8). Under M3 the screen resolves four tiers in shoot (42, 126, 61 and 6 accessions); under plant noise it resolves eight.

Resolution depth (Figure 5D). Under M3 the canonical construction resolves four tiers in shoot and five in root; under plant noise, eight and ten. Tier assignments of the constructions in use agree with the canonical tiers only moderately (Spearman 0.44–0.82, lowest for percentage increase); rescue and the canonical construction agree at 0.88–0.99. The practical reading is that this screen sorts accessions into a handful of classes, not into 235 ranks.

4.6 Batch

Col-0 was sown as an internal control every ~ 119 rows, which gives 13 usable blocks of 7 plants (Figure 2A). One-way analysis of variance rejects a common block mean:

- shoot: $F = 3.76$, $p = 1.7 \times 10^{-4}$;
- root: $F = 3.12$, $p = 1.2 \times 10^{-3}$.

The intraclass correlations [28] are 0.28 and 0.23. The between-block standard deviation is 0.43 (shoot) and 0.53 (root) of the between-accession standard deviation. By method of moments, $c_b = 0.081$ and 0.093 per cell. The three lines grown twice agree with this: their two block means differ by 19–42% (9807: 27.7 against 19.5 mg shoot).

The screen has no block term, so every accession’s value carries its block’s offset. Because Col-0 was repeated, the offsets are estimable for the inoculated–drought cell once block membership is known. The design already contains the control needed to remove the largest source of noise it has.

4.7 Collection site

Canonical shoot rescue declines towards the east ($\rho = -0.25$ with longitude, $p = 1.2 \times 10^{-4}$) and weakly towards the north ($\rho = -0.14$, $p = 0.037$). Vigour M_N increases in both directions ($\rho = 0.15$ and 0.18). Mock drought loss shows no geographic pattern in shoot ($p > 0.2$).

We ran 20 geographic tests. Two survive Bonferroni correction ($0.05/20 = 0.0025$): shoot rescue with longitude, and root vigour with longitude ($\rho = 0.20$, $p = 0.0024$). Root rescue does not ($p = 0.058$). The accessions were not sampled to be representative, so this is descriptive. A climate covariate from the collection coordinates [29] is the proper next step; it would turn the non-random composition of the panel from a reviewer’s objection into a measured variable.

5 Consequences for the screen

5.1 Triage of the candidate list

Of the 40 candidate rows, 28 genes, none appears in more than one trait family (Figure 6A). The seven genes that appear under several metrics do so only

across near-duplicates of one trait: average and median of the same trait, or shoot and total, which contains shoot. The candidates fall into four groups:

- *Mock-only (3 genes)*. MORN4 (AT1G77660), the antisense transcript AT2G07213 and TPS21 (AT5G23960) were found with a trait computed without inoculated plants. They are candidates for drought sensitivity, not for rescue.
- *Percentage increase (12 genes)*. These were found with traits whose rankings correlate at 0.78–0.81 with drought loss, and are at risk of being drought-sensitivity loci.
- *Rescue-derived (9 genes)*. These are the most interpretable candidates for the mutant screen.
- *Size and gain (4 genes)*. These carry the size loading.

This triage says nothing about whether a locus is real. It says which phenotype it is a locus *for*.

5.2 Checking mutant claims

The experimenter reports that mutants in plant iron-uptake genes show *higher* rescue. That fits a model in which the bacterium supplies iron independently of the plant [4; 6]. Figure 6B shows why the claim needs one extra number. Suppose a mutant only grows worse under drought (M_D reduced by ε) and receives the same absolute benefit. Then the constructions report:

- percentage increase: $1/(1 - \varepsilon)$ times more rescue, up to $2\times$ at $\varepsilon = 0.5$;
- gain: no change;
- rescue: a decrease, to $0.88\times$ at $\varepsilon = 0.5$;
- the canonical construction: a larger decrease, to $0.71\times$.

The claim of enhanced rescue is established if the absolute gain and the loss-normalised rescue both rise in the mutants. It is an artefact of the mutants’ weaker baseline if only the percentage increase does.

5.3 Design

Batch, not plant number, limits this screen (Figure 2D, Figure 6C). For shoot canonical rescue, reliability is:

- 0.34 at 7 plants in one run (the current design);
- 0.36 at 14 plants in one run;
- 0.50 at 7 plants in two runs, and 0.61 in three.

For root, reliability goes from 0.57 to 0.80 with three runs. Certifying equivalence at a margin of one true standard deviation becomes possible (probability 0.49) only for root with 28 plants in six runs, and it is not reached for shoot on this grid. Three recommendations follow:

- Repeat the screen in independent runs rather than adding plants.

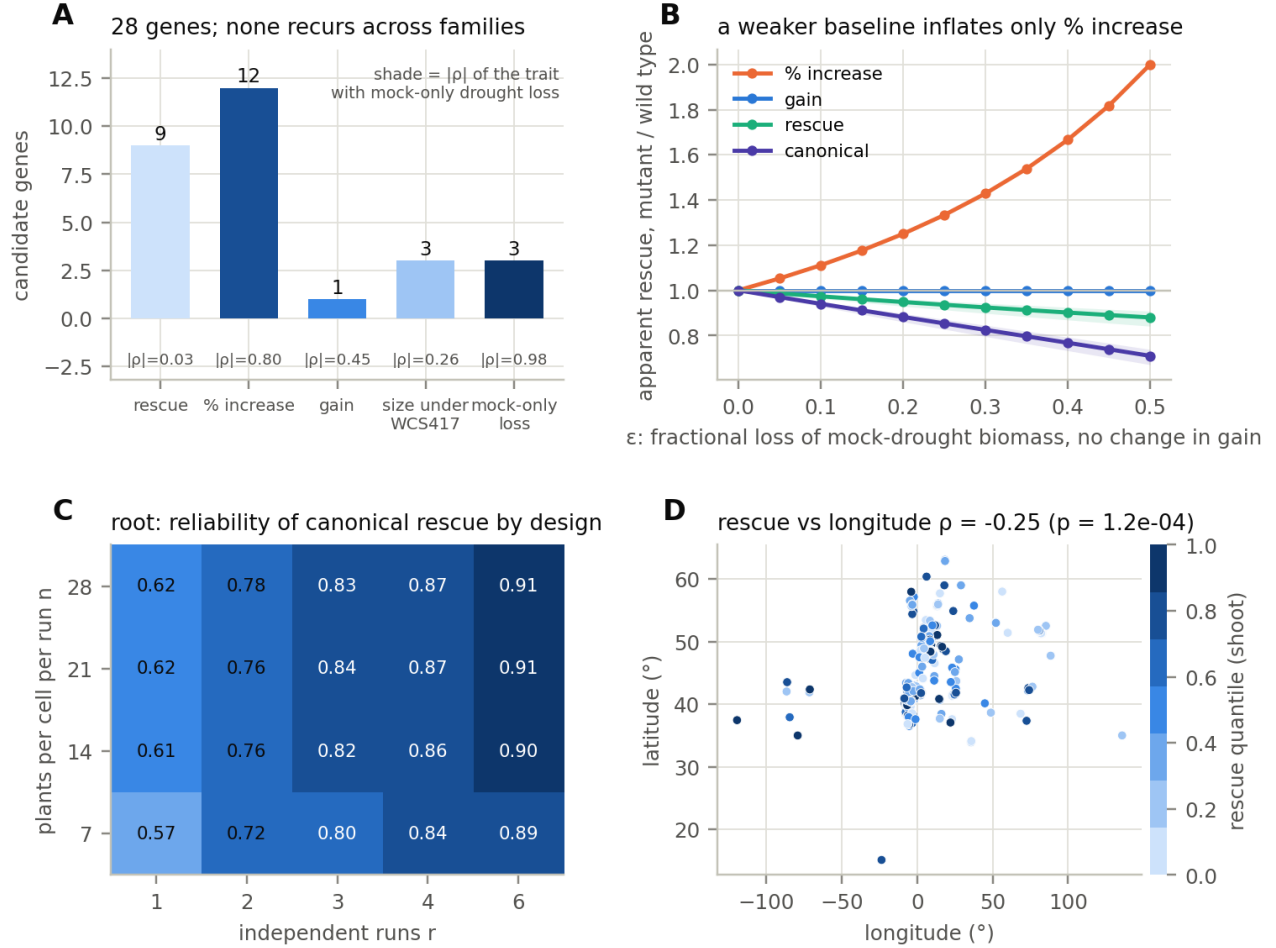


Figure 6: Consequences: candidate triage, a check for mutant claims, a replication design, and geography. (A) The 28 candidate genes by the trait family they were found in. Shading is the median $|\rho|$ between that family's traits and mock-only drought loss. No gene recurs across families. Three come from the mock-only family and twelve from percentage increase ($|\rho| = 0.80$). (B) Apparent change in rescue for a hypothetical mutant whose mock-drought biomass falls by a fraction ϵ while its absolute gain and non-stress biomass are unchanged. The median over accessions is shown, with interquartile band. Percentage increase reports up to a two-fold "enhancement"; gain reports none; rescue and the canonical construction report a decrease. (C) Reliability of root canonical rescue by plants per cell n and independent runs r . (D) Collection sites coloured by the quantile of canonical shoot rescue ($\rho = -0.25$ with longitude).

- (ii) Grow the mock and inoculated plants of an accession together, so that batch cancels in the gain.
- (iii) Keep a common control in every block and enter block as a term.

5.4 The analysis the data now support

Before any further association scan:

- (1) fix the phenotype to the canonical construction per organ, estimated as in §3.6;
- (2) fit a single model on per-plant data from all cells, with the inoculation $\times$ drought interaction as the phenotype and block as a term;
- (3) enter kinship [30; 31] and, for root, $\log M_N$ as co-variables;
- (4) report the resolution depth and the fraction of declined pairs alongside any ranking.

The first step needs no new data. The second needs the per-plant mock and non-stress data, which exist in the laboratory but not in the workbook.

6 Discussion

Rescue as a relation. The opening observation is simple. What an accession “does” with WCS417 under drought is not a property the accession carries. It is a relation among four cells, only one of which is directly the inoculated plant. A number assigned to the accession is a reading of that relation. Several readings are admissible and they disagree, and the disagreement is not noise (§4.3). This is the familiar statement that gene-by-environment interaction is the phenotype [15; 16], made operational. The object to be mapped is fixed only when a reading is fixed, and the reading should be fixed by the data.

Choice of reading as a testable hypothesis. The λ family turns “which formula?” into “which mechanism?” (Proposition 3.2): a multiplicative benefit on stressed biomass, or a restored fraction of the loss. The data answer that the benefit scales with the loss and not with the stressed biomass. That is a small biological result in its own right: WCS417 behaves as if it restores a share of what drought removes. It also has a methodological consequence. The trait that yielded twelve of the 28 candidates is the reading the data reject.

Comparison through four columns. Comparing accessions by their response alone treats two different things as one: a difference in what the bacterium does, and a difference in the plant it does it to. Carrying the mock baseline as two further columns separates them. The separation matters most where it is least visible, in false friends. These are pairs indistinguishable without the bacterium and different with it, and they are the contrasts a follow-up experiment should use.

Declining to decide. The third verdict value does work that no score can. On this screen most pairs *decline* (Figure 5C), and *correspond* is almost unreachable (Figure 5B). A method obliged to return a similarity for every pair would have returned 27,495 numbers, and nothing in them would have said that most of those numbers could not support a decision. We regard the minimum certifiable margin as the single most useful summary of a screen of this kind. It says how fine a distinction the experiment can support before any locus is sought.

The absence of loss of rescue. The screen could have detected complete loss in most backgrounds, and did not see it. That is consistent with the experimenter’s redundancy hypothesis, in which rescue runs through several routes and removing one leaves the others. It is also consistent with the iron result: up-take mutants are rescued more, not less. Under redundancy, the informative next experiment is combinatorial. In a network built from the time-resolved transcriptome, one would ask for the smallest set of genes whose joint removal disconnects the bacterial signal from the growth outcome, and test that set as a higher-order mutant. We have not done this; it requires the transcriptome.

7 Limitations

L1. Three cells have modelled, not measured, noise. The mock and non-stress spreads are assumed equal in relative terms to the inoculated-drought spread (Assumption 3.1). M4 (mock noise $\times 1.5$) changes no qualitative conclusion, but per-plant data for those cells would replace the assumption with a measurement.

L2. Batch is estimated from one cell and one genotype. Col-0 blocks exist only for inoculated-drought. We assumed the same batch coefficient of variation elsewhere and independent batch effects across cells; shared trays would make M3 pessimistic.

L3. No genotypes. We did not rerun the association, correct it for kinship or population structure, or test any candidate.

L4. One run. Every conclusion about construction dependence could be sharpened, or reversed in shoot, by a second run. The design analysis shows that a second run is the most valuable single addition.

L5. The λ model is a model. (5) is one family of mechanisms. The root rejection of $b_1 + b_2 = 1$ shows that it is incomplete there. The simulation-based bias correction assumes the noise model.

L6. Margins are choices. The verdicts depend on δ . We report sweeps and the minimum certifiable

margin, which do not depend on a choice, but the four-column table does.

L7. W_N is unidentified. Constructions that use the inoculated non-stress plants, for example a full interaction contrast, cannot be evaluated from the workbook.

8 Conclusion

A screen for rhizobacterial drought rescue measures a relation, and its derived traits are readings of that relation. On 235 *Arabidopsis* accessions inoculated with WCS417, the readings in use disagree beyond noise in root, and under plant-level noise they certify hundreds of pairs in opposite order. Percentage increase, the source of the largest group of candidate loci, is the reading the data reject, because it largely ranks drought sensitivity. The data select a loss-normalised rescue. Within the admissible range the choice stops mattering more than noise. Compared through four columns with a calibrated three-valued verdict, the screen certifies divergence but almost never equivalence, and resolves four to five tiers of rescue. Its largest source of noise is batch, which its own Col-0 control can measure and remove. The next steps are the phenotype fixed before mapping, the per-plant model with block and kinship, a second independent run, and a retest of Per-1. None requires a new idea, only the decision to let the data choose the phenotype and the discipline to report when they cannot decide.

Data and code availability

The screen’s data are unpublished and belong to the laboratory that generated them. The validation suite (`validation/`) regenerates every number in this paper from the workbook, deterministically, from seed 417, in about ten seconds. Results are written to `results/` as one JSON file per experiment (E1–E12), and figures are rebuilt from those files by `validation/make_figures.py`. Analyses use NumPy, SciPy, pandas, scikit-learn and Matplotlib [32; 33; 34; 35; 36]; the accompanying interactive specification uses D3 [37].

Acknowledgements

We thank the laboratory that designed and ran the screen, generated all data analysed here, and described the experiment in detail.

Conflicts of interest

None declared.

References

- [1] Corn   M. J. Pieterse, Christos Zamioudis, Roeland L. Berendsen, David M. Weller, Saskia C. M. Van Wees, and Peter A. H. M. Bakker. Induced systemic resistance by beneficial microbes. *Annual Review of Phytopathology*, 52:347–375, 2014.
- [2] Christos Zamioudis, Parthena Mastranesti, Pankaj Dhonukshe, Ikram Blilou, and Corn   M. J. Pieterse. Unraveling root developmental programs initiated by beneficial *Pseudomonas* spp. bacteria. *Plant Physiology*, 162(1):304–318, 2013.
- [3] Roeland L. Berendsen, Marcel C. van Verk, Ioannis A. Stringlis, Christos Zamioudis, Jan Tommassen, Corn   M. J. Pieterse, and Peter A. H. M. Bakker. Unearthing the genomes of plant-beneficial *Pseudomonas* model strains WCS358, WCS374 and WCS417. *BMC Genomics*, 16:539, 2015.
- [4] Christos Zamioudis, Jolanda Korteland, Johan A. Van Pelt, Muriel van Hamersveld, Nina Dombrowski, Yang Bai, Johannes Hanson, Marcel C. Van Verk, Hong-Qing Ling, Paul Schulze-Lefert, and Corn   M. J. Pieterse. Rhizobacterial volatiles and photosynthesis-related signals coordinate *MYB72* expression in *Arabidopsis* roots during onset of induced systemic resistance and iron-deficiency responses. *The Plant Journal*, 84(2):309–322, 2015.
- [5] Ioannis A. Stringlis, Ke Yu, Kirstin Feussner, Ronnie de Jonge, Sietske Van Bentum, Marcel C. Van Verk, Roeland L. Berendsen, Peter A. H. M. Bakker, Ivo Feussner, and Corn   M. J. Pieterse. MYB72-dependent coumarin exudation shapes root microbiome assembly to promote plant health. *Proceedings of the National Academy of Sciences USA*, 115(22):E5213–E5222, 2018.
- [6] Eline H. Verbon, Pauline L. Trapet, Ioannis A. Stringlis, Sophie Kruijs, Peter A. H. M. Bakker, and Corn   M. J. Pieterse. Iron and immunity. *Annual Review of Phytopathology*, 55:355–375, 2017.
- [7] Paul C. A. Wintermans, Peter A. H. M. Bakker, and Corn   M. J. Pieterse. Natural genetic variation in *Arabidopsis* for responsiveness to plant growth-promoting rhizobacteria. *Plant Molecular Biology*, 90(6):623–634, 2016.
- [8] Franciska T. de Vries, Rob I. Griffiths, Christopher G. Knight, Oceane Nicolitch, and Alex Williams. Harnessing rhizosphere microbiomes for drought-resilient crop production. *Science*, 368(6488):270–274, 2020.
- [9] Susanna Atwell, Yu S. Huang, Bjarni J. Vilh  lms  n, Glenda Willems, Matthew Horton, Yan Li, Dazhe Meng, Alexander Platt, Aaron M. Tarone, Tina T. Hu, et al. Genome-wide association study of 107 phenotypes in *Arabidopsis thaliana* inbred lines. *Nature*, 465:627–631, 2010.

- [10] Dominik G. Grimm, Damián Roqueiro, Patrice A. Salomé, Stefan Kleeberger, Bastian Greshake, Wangsheng Zhu, Chang Liu, Christoph Lippert, Oliver Stegle, Bernhard Schölkopf, Detlef Weigel, and Karsten M. Borgwardt. easyGWAS: A cloud-based platform for comparing the results of genome-wide association studies. *The Plant Cell*, 29(1):5–19, 2017.
- [11] Karl Pearson. Mathematical contributions to the theory of evolution.—on a form of spurious correlation which may arise when indices are used in the measurement of organs. *Proceedings of the Royal Society of London*, 60:489–498, 1897.
- [12] Richard A. Kronmal. Spurious correlation and the fallacy of the ratio standard revisited. *Journal of the Royal Statistical Society, Series A*, 156(3): 379–392, 1993.
- [13] R. A. Fischer and R. Maurer. Drought resistance in spring wheat cultivars. I. Grain yield responses. *Australian Journal of Agricultural Research*, 29(5):897–912, 1978.
- [14] Paul E. Verslues, Manu Agarwal, Surekha Katiyar-Agarwal, Jianhua Zhu, and Jian-Kang Zhu. Methods and concepts in quantifying resistance to drought, salt and freezing, abiotic stresses that affect plant water status. *The Plant Journal*, 45(4):523–539, 2006.
- [15] Mohamed El-Soda, Marcos Malosetti, Bas J. Zwaan, Maarten Koornneef, and Mark G. M. Aarts. Genotype $\times$ environment interaction QTL mapping in plants: lessons from *Arabidopsis*. *Trends in Plant Science*, 19(6):390–398, 2014.
- [16] David L. Des Marais, Kyle M. Hernandez, and Thomas E. Juenger. Genotype-by-environment interaction and plasticity: exploring genomic responses of plants to the abiotic environment. *Annual Review of Ecology, Evolution, and Systematics*, 44:5–29, 2013.
- [17] 1001 Genomes Consortium. 1,135 genomes reveal the global pattern of polymorphism in *Arabidopsis thaliana*. *Cell*, 166(2):481–491, 2016.
- [18] Raymond J. Carroll, David Ruppert, Leonard A. Stefanski, and Ciprian M. Crainiceanu. *Measurement Error in Nonlinear Models: A Modern Perspective*. Chapman & Hall/CRC, Boca Raton, 2 edition, 2006.
- [19] William G. Cochran. The combination of estimates from different experiments. *Biometrics*, 10(1):101–129, 1954.
- [20] Julian P. T. Higgins and Simon G. Thompson. Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, 21(11):1539–1558, 2002.
- [21] Douglas S. Falconer and Trudy F. C. Mackay. *Introduction to Quantitative Genetics*. Longman, Harlow, 4 edition, 1996.
- [22] Michael Lynch and Bruce Walsh. *Genetics and Analysis of Quantitative Traits*. Sinauer, Sunderland, MA, 1998.
- [23] Donald J. Schuirmann. A comparison of the two one-sided tests procedure and the power approach for assessing the equivalence of average bioavailability. *Journal of Pharmacokinetics and Biopharmaceutics*, 15(6):657–680, 1987.
- [24] Erich L. Lehmann and Joseph P. Romano. *Testing Statistical Hypotheses*. Springer, New York, 3 edition, 2005.
- [25] Bradley Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1): 1–26, 1979.
- [26] Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- [27] Paul Jaccard. The distribution of the flora in the alpine zone. *New Phytologist*, 11(2):37–50, 1912.
- [28] Patrick E. Shrout and Joseph L. Fleiss. Intraclass correlations: uses in assessing rater reliability. *Psychological Bulletin*, 86(2):420–428, 1979.
- [29] Stephen E. Fick and Robert J. Hijmans. WorldClim 2: new 1-km spatial resolution climate surfaces for global land areas. *International Journal of Climatology*, 37(12):4302–4315, 2017.
- [30] Jianming Yu, Gael Pressoir, William H. Briggs, Irie Vroh Bi, Masanori Yamasaki, John F. Doebley, Michael D. McMullen, Brandon S. Gaut, Dahlia M. Nielsen, James B. Holland, Stephen Kresovich, and Edward S. Buckler. A unified mixed-model method for association mapping that accounts for multiple levels of relatedness. *Nature Genetics*, 38:203–208, 2006.
- [31] Hyun Min Kang, Jae Hoon Sul, Susan K. Service, Noah A. Zaitlen, Sit-ye Kong, Nelson B. Freimer, Chiara Sabatti, and Eleazar Eskin. Variance component model to account for sample structure in genome-wide association studies. *Nature Genetics*, 42:348–354, 2010.
- [32] Charles R. Harris, K. Jarrod Millman, Stéfan J. van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, et al. Array programming with NumPy. *Nature*, 585:357–362, 2020.
- [33] Pauli Virtanen, Ralf Gommers, Travis E. Oliphant, Matt Haberland, Tyler Reddy, David Cournapeau, et al. SciPy 1.0: fundamental algorithms for scientific computing in Python. *Nature Methods*, 17:261–272, 2020.

- [34] Wes McKinney. Data structures for statistical computing in Python. In *Proceedings of the 9th Python in Science Conference*, pages 56–61, 2010.
- [35] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, et al. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [36] John D. Hunter. Matplotlib: A 2D graphics environment. *Computing in Science & Engineering*, 9(3):90–95, 2007.
- [37] Michael Bostock, Vadim Ogievetsky, and Jeffrey Heer. D³: Data-driven documents. *IEEE Transactions on Visualization and Computer Graphics*, 17(12):2301–2309, 2011.